

# Final Engineering Design Report (FEDR)

## Purpose of the FEDR

The Final Engineering Design Report (FEDR) is the capstone written submission for the MSc Group Design Project. Its aim is to present a clear, coherent, and well-reasoned account of the development of your aircraft concept, showing:

- The final design and why it was selected
- How modelling choices, hyperparameters, and trade-offs shaped the design
- How discipline-level tools integrate into a system-level MDO framework
- What the team learned: technically, organisationally, and through iteration
- What uncertainties remain, and how a future engineering team should progress

The FEDR is not a “*data dump*” of everything you did. It is a strategic, insight-driven technical document that demonstrates engineering judgement.

## What the FEDR Should Cover

Your report should follow the mandatory chapter and section structure provided in the “*Table of Contents for the FEDR*” document (supplied separately). This structure ensures consistent assessment across groups and reflects the expected flow of engineering reasoning.

The table of contents document provides suggested page counts for each section of the report. Chapter page limits will not be enforced; however, there is a strict **66-page limit** for the entire document (excluding references and Appendices). *Teams that exceed this limit may be subject to a 1-point deduction per additional page*

## Principles of a Strong FEDR

### Sell your knowledge, not your Design

Your primary aim is to demonstrate what your team learned as you developed and integrated your methods, showing that your conclusions reflect sound engineering judgement.

For example, the assessor should see that you have understood your modelling choices, applied appropriate fidelity, and recognised key uncertainties.

The FEDR should emphasise the engineering logic behind the design, not read like a brochure for the aircraft.

### Lead with conclusions, follow with analysis

Writing a design report is different from a lab report or academic paper because industry practice expects key results to appear before detailed analysis. In a professional environment, your boss might read only the executive summary and introduction, and their boss might read only the executive summary.

The analysis is assumed correct once the report has passed up the chain. For this reason, the FEDR should present conclusions early so the reader understands the essential message before engaging with the supporting technical detail.

### Prioritise critical reflections and clarity over completeness

Markers are not specialists in every discipline. They need to understand what modelling choices matter for your concept, what simplifications you made and why, and where your model is strong (but more appropriately, where it is weak).

Do not include derivations or large tables of intermediate results. Those belong in appendices (which are not assessed).

### Use figures, diagrams, and “walk-around” visuals

Well-designed diagrams often communicate system-level logic better than text. Where possible, try to incorporate figures and illustrations to help convey your understanding. However, avoid overly busy visuals, which detract from the narrative. Figures should help guide the reader through the story.

### Focus on meaningful sensitivity and trade studies

A strong FEDR will not simply present a sequence of trade studies; it will use them to interpret influential parameters and unexpected interdisciplinary interactions. It may then show how these sensitivities informed engineering design decisions.

### Treat Project Management as a real engineering discipline

Your project management section should show how quantitative evidence (such as progress metrics, integration data, or risk indicators) was used to guide decisions and refine workflow practices, rather than relying on subjective or poorly substantiated decision-making. Present your management choices as deliberate, data-driven adjustments that improved coordination and strengthened the overall technical process.

### Write for the assessor, not for your team

Write for an aerospace engineer who understands the fundamentals of aerospace engineering but is not a specialist in conceptual design. The assessor will not share your team’s deep knowledge of your concept, so use clear language, define acronyms, and ensure figures are self-contained. Aim for a concise narrative that communicates your reasoning to a general aerospace audience without relying on specialist assumptions.

## Uploading Your FEDR

Ensure you upload a PDF document with a maximum file size of 10 MB.

## Final Advice

A successful FEDR demonstrates clear engineering judgement, a well-argued, evidence-based design history, and an honest reflection on the methodologies, assumptions, and integration developed. To achieve this, it is important to:

- **Be clear rather than comprehensive** – we’re not looking for a dump of all that you’ve learnt, we want you to demonstrate a good understanding of the interdisciplinary nature of aircraft design.
- **Avoid unnecessary detail** – heavy derivations and raw simulation outputs, if required, should be placed in an appendix. Ask yourself: *Is this section worth the space on the page?* Your maximum page count is strict; prioritise insight over volume.
- **State your assumptions** – a successful team will know the limitations of their modelling assumptions
- **Use simple diagrams and sketches where possible** – help the panel visualise what you are talking about
- **Keep things consistent** - use consistent units, fonts, colours<sup>1</sup> and figure styles. Try to use a single coherent voice.

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<sup>1</sup> Approximately 1 in 8 men and 1 in 200 women are colourblind. Using a colourblind safe colour palette ensures your report is accessible to all readers. (<https://doi.org/10.1038/d41586-021-02696-z>)